KOOLIE : BRIDGING THE GAP IN LOCAL SERVICES

MINI PROJECT REPORT IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled "**KOOLIE - BRIDGING THE GAP IN LOCAL SERVICES**", submitted by **ANFAS RAHMAN T R (TLY23CS026),BENEDICT JOSEPH (TLY23CS039), JASNA JALEEL (TLY23CS062), ADITHYAN BABU(LTLY23CS135)** in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering to APJ Abdul Kalam Technological University, Kerala**, is a record of bona fide work carried out under my guidance and supervision

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- To train youngsters to deploy technology for the service of humanity.

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3. Promote the design, research, and implementation of innovative products and services in Computer Science & Engineering, utilizing strong technical, communication, leadership, and entrepreneurial skills.

DECLARATION

We hereby declare that the project report titled **"KOOLIE : BRIDGING THE GAP IN LOCAL SERVICES"** is an original and bonafide record of the work carried out by our team in partial fulfilment of the requirements for the award of the **Bachelor of Technology degree in Computer Science and Engineering under APJ Abdul Kalam Technological University, Kerala**. We affirm that this report embodies our genuine efforts and adheres to the highest standards of academic integrity. All data, references, and sources have been properly acknowledged, and no part of this work has been misrepresented or plagiarized. We understand that any violation of these principles may result in disciplinary action by the institution or University. Furthermore, we certify that this report has not been submitted to any other university, institute, or organization for the award of any degree, diploma, or similar qualification.

Place: Thalassery

Date : 06 /04/2026

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ABSTRACT

Koolie is a digital platform designed to bridge the gap between service seekers and local service professionals by providing a centralized system for managing household and local services. The platform enables users to post service requests for tasks such as plumbing, electrical work, carpentry, and general maintenance, while service professionals can register, browse available requests in their area, and accept jobs based on their skills and availability. Koolie also integrates an in-app communication feature that allows direct interaction between clients and professionals, ensuring better coordination, clarity of requirements, and efficient service delivery. The system is developed using React.js for the frontend and Firebase for backend services and database management, which supports real-time data synchronization and smooth performance. It focuses on delivering a user-friendly interface that simplifies service booking and enhances accessibility for all users. Additionally, the platform improves transparency by systematically managing service requests and interactions. By reducing dependency on informal methods such as word-of-mouth, Koolie provides a more organized and reliable solution for local service management. Overall, the platform not only simplifies the process of finding trusted service providers but also creates better employment opportunities for skilled professionals, contributing to a more efficient and structured local service ecosystem

CONTENTS

CERTIFICATE.....II

VISION AND MISSION.....III

DECLARATION.....V

ACKNOWLEDGMENT.....VI

ABSTRACT.....VII

CONTENTS.....VIII

LIST OF FIGURES.....X

1. INTRODUCTION.....1

1.1 Objectives.....2

1.2 Methodology.....2

2. LITERATURE REVIEW.....3

3. PROBLEM STATEMENT.....6

4. EXISTING SYSTEM.....7

5. PROPOSED SYSTEM.....8

6. SOFTWARE AND HARDWARE REQUIREMENTS.....9

7. DEVELOPMENT TOOLS.....10

7.1 Integrated Development Environment (IDE).....10

7.2 Application Development Tools.....10

7.3 Version Control System.....11

8. SYSTEM DESIGN.....12

8.1 System Architecture.....12

8.2 Use case Diagram.....13

8.3 Data Flow Diagram.....14

9. MODULAR DIVISIONS.....16

9.1 Admin Module.....16

9.2 User Module.....16

9.3 Service Provider Module.....16

10. CODING PROCEDURE.....17

10.1 Front End (React).....17

10.2 Back End (Firebase).....17

10.3 Session Management.....18

10.4 Admin Module.....18

11. TESTING AND IMPLEMENTATION.....19

12. RESULTS AND DISCUSSION.....27

13. CONCLUSION AND FUTURE SCOPE.....28

14. BIBLIOGRAPHY.....29

LIST OF FIGURES

Figure 8.1: System Architecture.....12
Figure 8.2: Use Case Diagram 13
Figure 8.3: Data Flow Diagram Level-0.....14
Figure 8.3: Data Flow Diagram Level-1 15
Figure 8.3: Data Flow Diagram Level-2.....15
Figure 11.1:Home page….....22
Figure 11.2: Signup page.....22
Figure 11.3 Client Dashboard.....23
Figure 11.4 Provider Dashboard.....24
Figure 11.5.1 :Code…….…….25
Figure 11.5.2 :Code…….…….25
Figure 11.5.3 :Code…….…….26

1. INTRODUCTION

In today's fast-paced world, accessing reliable local service professionals has become a common challenge for many individuals. People often require services such as plumbing, electrical work, carpentry, cleaning, and other household maintenance tasks. However, finding trustworthy and skilled professionals nearby is often difficult due to the lack of a centralized system. Most users still depend on word-of-mouth recommendations, personal contacts, or unorganized sources, which can be time-consuming and unreliable. At the same time, many skilled service professionals face challenges in finding consistent job opportunities. Due to limited visibility and lack of digital presence, they often miss potential work despite having the required skills. This creates a gap between service seekers and service providers, resulting in inefficiency in service delivery and reduced employment opportunities for local workers.

To address these issues, the Koolie platform is developed as a digital solution that connects service seekers with local service professionals. Koolie acts as a centralized platform where users can easily post service requests based on their needs. These requests are then made available to registered professionals who can browse and accept jobs according to their skills and availability. This system simplifies the process of finding and hiring service providers while ensuring better accessibility and efficiency. One of the key features of the Koolie platform is its in-app communication system, which allows direct interaction between clients and professionals. This feature enables users to discuss service details such as requirements, timing, and pricing before confirming the job, thereby improving clarity and reducing misunderstandings. The platform also ensures a structured workflow for service requests, making the overall process more organized and userfriendly. Koolie is developed using modern web technologies such as React.js for the frontend and Firebase for backend services and database management. These technologies enable real-time updates, efficient data handling, and a smooth user experience. The platform is designed to be simple, responsive, and accessible to users with varying levels of technical knowledge.

Overall, Koolie aims to bridge the gap between service seekers and professionals by providing a reliable, efficient, and easy-to-use platform. It not only helps users find trusted local services quickly but also creates better job opportunities for skilled workers within the community.

By digitizing and organizing the local service ecosystem, Koolie contributes to improving service accessibility and supporting the growth of local employment.

1.1 Objectives

- To simplify the process of finding reliable and trusted local service providers.
- To improve service coordination and reduce time required for service completion.
- To increase visibility and job opportunities for skilled local workers.

1.2 Methodology

The methodology of the Koolie platform begins with analyzing the existing system of finding local service professionals to identify its limitations such as lack of organization, poor communication, and dependency on informal methods. Based on this analysis, the system is designed to provide a centralized digital platform that connects service seekers with service professionals.Users can register on the platform and submit service requests by specifying their requirements. These requests are stored securely in the database and made available to registered service professionals. Professionals can browse the available requests and accept jobs based on their skills, location, and availability.

The platform also implements an in-app communication system that allows direct interaction between clients and professionals for better coordination and clarity of service details. Rolebased access is maintained to ensure that users and professionals can only access relevant features and data.Finally, the system supports efficient service management by enabling smooth request handling, communication, and job completion. Continuous monitoring and improvements are carried out to enhance user experience and ensure reliable service delivery

2.LITERATURE REVIEW

The growth of digital service platforms has increased the need for efficient systems that connect service seekers with local service professionals. Traditional methods of finding services, such as relying on personal contacts or word-of-mouth, are often unorganized, timeconsuming, and unreliable.This creates challenges in accessing trusted and timely services.

Existing research focuses on improving service accessibility, user experience, and providing better opportunities for professionals through digital platforms. However, many systems lack effective communication features and simple user-friendly design. This highlights the need for a platform like Koolie, which aims to provide a centralized and efficient solution for local service management.

2.1 UrbanClap: India's Largest Home Service Provider – A Case Study

Sharma and Mehta analyzed the working model of Urban Company (formerly UrbanClap), one of the largest home service platforms in India. Their study highlighted how digital platforms efficiently connect service seekers with verified professionals and manage service requests through a centralized system. The research emphasized the importance of service quality, professional verification, and trust in improving customer satisfaction and platform reliability. The findings demonstrate how a structured digital marketplace can simplify service accessibility. This study supports the development of a centralized service platform like Koolie, which aims to provide reliable and efficient service connections between users and professionals.

2.2 Examining Consumers' Behavioral Intentions Towards Online Home Service Applications

Jain and Agarwal examined the factors influencing consumer adoption of online home service applications. Their study identified several key determinants, including ease of use, perceived trust, and service reliability, which significantly affect user satisfaction and continued platform usage. The research highlighted that a well-designed and user-friendly interface plays a major role in encouraging customers to adopt digital service platforms. These findings

emphasize the importance of designing an intuitive and reliable system, which aligns with the objectives of the Koolie platform in providing a simple and efficient user experience.

2.3 The Impact of On-Demand Home Service Apps on the Gig Economy

Suresh and Anitha explored the impact of on-demand home service applications on the growth of the gig economy. Their research demonstrated that digital service platforms create new employment opportunities for skilled professionals by enabling them to offer services independently through online marketplaces. At the same time, such platforms improve service accessibility for customers by providing quick and convenient booking options. The study indicates that digital platforms play an important role in bridging the gap between service providers and users. This research supports the concept that the Koolie platform can contribute to local employment generation while efficiently connecting service providers with service seekers.

2.4 Service Quality and Customer Satisfaction in Online Home Service Platforms

Karthikeyan and Priyanka investigated the relationship between service quality and customer satisfaction in online service platforms. Their study identified critical factors such as response time, pricing transparency, service consistency, and communication efficiency as key determinants of user satisfaction. The research emphasized that maintaining high service quality standards is essential for retaining customers and building trust in digital service platforms. These findings highlight the importance of efficient service coordination and clear communication features, which are integrated into the Koolie platform to ensure reliable and satisfactory service delivery.

TITLE	AUTHOR	YEAR	ADVANTAGES	DISADVANTAGES
UrbanClap: India's Largest Home Service Provider – A Case Study	Nivedita Sharma, Rahul Mehta	2022	Provides real-world insights into large-scale service platforms	Focuses more on business aspects than system design
Platform Business Models and Service Innovation in the Digital Economy	N. Kumar, A. Prakash, M. Arora	2023	Examines how digital platforms drive service innovation and business transformation.	Broad analysis with limited system-level technical insights.
Platform-Based Approaches in the AEC Industry: A Bibliometric Review and Trend Analysis	Layla Mujahed, Gang Feng, Jianghua Wang	2026	Provides trend analysis of platform-based systems and explains how modular platform structures improve efficiency and scalability in digital ecosystems.	Focuses mainly on the construction industry rather than general service marketplaces.
Examining Consumers' Behavioral Intentions Towards Online Home Services Applications	Shweta Jain, Ankit Agarwal	2023	Identifies key factors influencing user adoption of home service applications	Does not address technical or system design aspects
An Empirical Study on Urban Company Service Quality Influence on Customers	**:** R. Karthikeyan, S. Priyanka	2023	Identifies gaps in pricing transparency and service consistency	Limited to a single service platform
Platform Design and Governance in Industrial Markets: Charting the MetaOrganizational Logic	Virginia Springer, Krithika Randhawa, Marin Jovanovic, Paavo Ritala, Frank T. Piller	2025	Introduces governance models for digital platforms and explains how platform architecture manages interactions between multiple stakeholders.	Focuses mainly on industrial B2B platforms rather than consumer service platforms.
The Impact of On-Demand Home Service Apps on the Gig Economy	R. Suresh, M. Anitha	2024	Highlights challenges faced by local service professionals	Does not address platform optimization techniques
Predicting Customer Satisfaction in Urban Company Using Machine Learning	A. Kumar, P. Singh	2022	Identifies key service parameters affecting user experience	Requires large datasets for accurate prediction

3. PROBLEM STATEMENT

In many local communities, finding reliable service professionals for household and maintenance tasks is still handled through informal methods such as word-of-mouth, personal contacts, or unorganized sources. This approach often leads to delays, lack of trust, and difficulty in accessing skilled workers. Additionally, service professionals face challenges in finding consistent job opportunities due to limited visibility. These issues highlight the need for a centralized and efficient digital solution.

1. Limited Job Opportunities for Professionals

Skilled workers may not get enough exposure to potential clients, reducing their chances of finding work.

2. Time-Consuming Coordination

Managing service requests, scheduling, and confirmations manually takes more time and effort.

3. Absence of Centralized Service Platform

Without a unified system, managing service requests and professional interactions becomes complex and Inefficient

These challenges emphasize the need for the Koolie platform, which provides a centralized, user-friendly system to connect service seekers with professionals, improve communication, and ensure efficient service management.

4. EXISTING SYSTEM

In the current scenario, clients primarily rely on existing service applications or personal/local contacts to find professionals such as plumbers, electricians, or technicians. While these methods offer some level of convenience, they come with several limitations. Most existing service platforms are designed with a strong focus on urban and metropolitan areas. As a result, users in smaller towns and rural regions often face difficulty accessing reliable and skilled professionals. This creates an imbalance in service availability and limits opportunities for both clients and service providers in less-developed areas.

Additionally, service providers working through these platforms frequently encounter high commission charges and strict platform policies. These factors reduce their earnings and restrict their flexibility in accepting or negotiating work. Many professionals are therefore hesitant to rely completely on such platforms. Another significant drawback is the limited scope for direct communication between clients and service providers before booking a service. Clients are often unable to clearly explain their requirements, and professionals may not fully understand the task in advance. This can lead to misunderstandings, inefficiencies, and dissatisfaction on both sides. Moreover, the existing systems offer limited customization options, especially for urgent services or location-specific needs. Clients with immediate requirements or unique situations may struggle to find suitable professionals quickly.

LIMITATIONS OF THE EXISTING SYSTEM

- Focus mainly on urban areas; limited access in rural and small towns
- Limited direct communication before booking
- Possibility of misunderstandings between clients and professionals
- Difficulty in finding services in specific or remote locations
- Dependence on centralized platforms
- Limited customization of services based on user needs

5.PROPOSED SYSTEM

The proposed system aims to address the limitations of the existing approach by introducing a centralized and flexible platform that connects clients with local service professionals more efficiently. This platform is designed to support professionals from all regions, including small towns and rural areas, thereby ensuring wider accessibility and inclusivity. By doing so, it not only helps clients find services easily but also creates more job opportunities for independent and local workers. One of the key advantages of the proposed system is improved job visibility for service providers. Professionals can showcase their skills, availability, and service areas, enabling them to reach a larger audience without being restricted by location or platform bias.

The system also reduces dependency on rigid service models and high commission structures. By offering a more balanced and fair approach, it empowers professionals to work independently while still benefiting from a digital platform.To enhance user experience, the platform enables direct and clear communication between clients and service providers before booking. This ensures better understanding of requirements, accurate pricing, and increased trust.

Furthermore, the proposed system emphasizes simplicity and user-friendliness. It is designed to provide a smooth and efficient experience for both clients and professionals, allowing quick service discovery, easy booking, and effective interaction.Overall, the proposed solution creates a more accessible, flexible, and efficient ecosystem for local service connections, benefiting both service seekers and providers.

KEY FEATURES:

- ✦ Centralized platform for connecting clients and service professionals
- ✦ Supports users from both urban and rural areas
- ✦ Enables direct communication between clients and professionals
- ✦ Provides better visibility for local and independent workers
- ✦ Reduces dependency on high-commission platforms
- ✦ Flexible pricing and service negotiation options
- ✦ Quick access to services, including urgent requests
- ✦ Location-based service matching
- ✦ Simple and user-friendly interface
- ✦ Efficient booking and service management system

6. SOFTWARE AND HARDWARE REQUIREMENTS

The hardware and software requirements for the installation and smooth functioning of Koolie may vary depending on the operating environment. This section describes the minimum configuration required for the proper execution and effective performance of the system.

6.1 SOFTWARE REQUIREMENTS

Software Framework	React ^19.2.0 + Vite ^7.2.4 (Single-Page Application)
Programming Language	JavaScript (JSX, ES2022+)
Backend	Firebase – Authentication, Cloud Firestore
Runtime Environment	Node.js v18.x or v20.x LTS
Key Development Tools	VS Code, Git

6.2 HARDWARE REQUIREMENTS

Processor	Intel i5 / AMD Ryzen 5 or higher
RAM	8 GB (recommended for emulators and indexing)
Storage	SSD with at least 1 GB free
End-User Requirements	Standard smartphone or computer with internet access

7. DEVELOPMENT TOOLS

7.1 Integrated Development Environment (IDE):

An Integrated Development Environment (IDE) provides a comprehensive platform for designing, developing, and testing software applications. For the development of Koolie, Visual Studio Code (VS Code) is used as the primary IDE. It offers powerful features such as source code editing, debugging support, extension integration, and version control compatibility.

VS Code supports React and JavaScript development natively and integrates seamlessly with Firebase, Vite, ESLint, and Git — making it an ideal choice for building the frontend and connecting to backend cloud services. The IDE helps in organising project files efficiently and streamlines the overall development workflow.

7.2 Application Development Tools :

Frontend — React + Vite

The frontend of Koolie is developed using React ^19.2.0, which enables the creation of responsive and interactive user interfaces through a component-based architecture. Vite ^7.2.4 serves as the build tool and development server, offering near-instant Hot Module Replacement (HMR) and optimised production bundling. React Router DOM ^7.13.1 manages all client-side navigation within the Single-Page Application.

React provides a consistent and reusable UI experience, allowing faster development from a single JavaScript codebase without requiring separate mobile or desktop builds.

Backend Technologies — Firebase

Firebase is used as the backend platform for Koolie, handling authentication, business logic, and real-time data synchronisation. Firebase services support seamless communication between the frontend React components and the cloud backend, eliminating the need for a custom server.

Database Management — Cloud Firestore

The system uses Cloud Firestore, a NoSQL document database provided by Firebase, for flexible and scalable data storage. Firestore stores all core application data including user profiles, authentication records, and any project-related information. Its real-time synchronisation ensures that updates are immediately reflected across all connected clients without requiring manual refresh. Firestore's document-collection model allows the application data to scale efficiently and supports complex querying without compromising performance.

7.3 Version Control System:

A version control system is essential for managing source code changes and supporting collaborative development. Git is used in Koolie to track code modifications, manage feature branches, and maintain consistency throughout the development lifecycle.

GitHub is used as the remote hosting platform for the project repository. It enables the team to manage versions, coordinate contributions, review code changes, and ensure smooth integration of new features. This approach simplifies debugging, supports rollback when needed, and keeps the codebase organised at every stage of development.

8. SYSTEM DESIGN

8.1 System Architecture

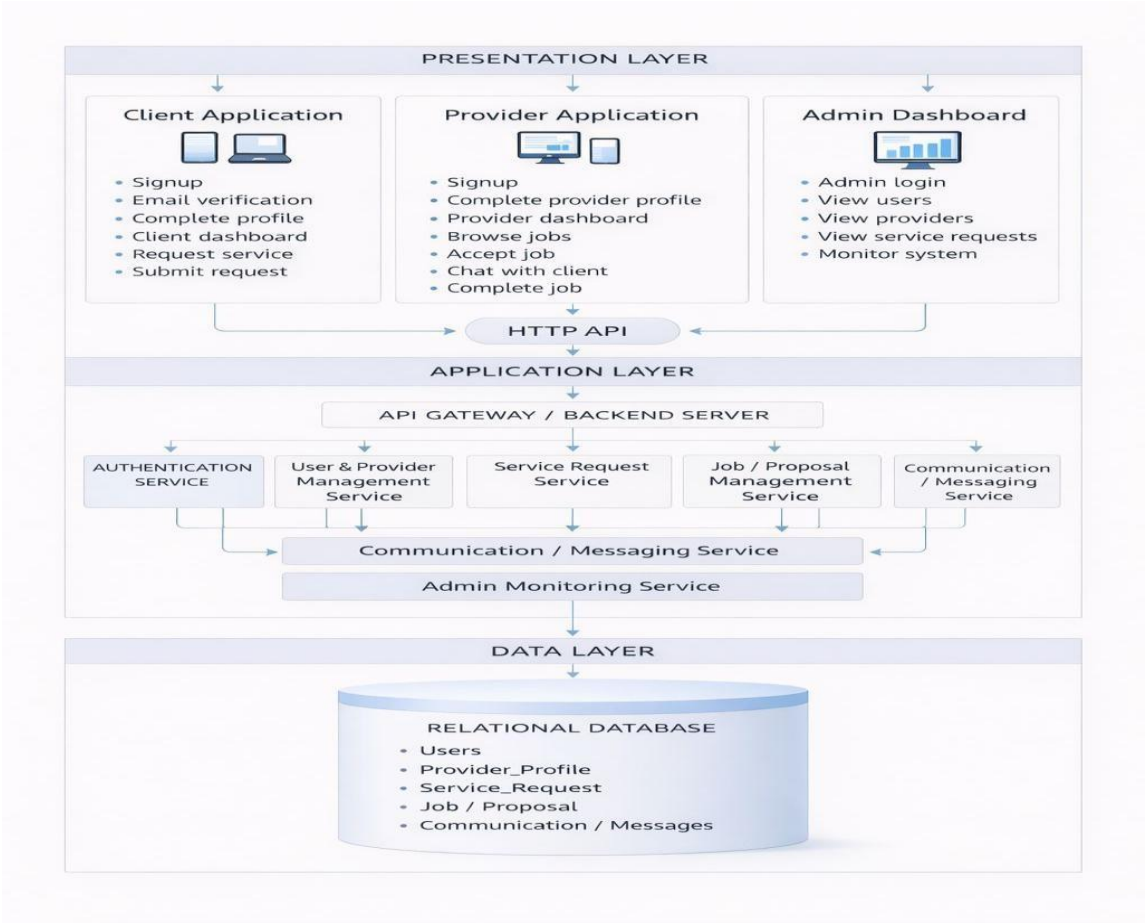


Figure 8.1

- **Client Module** Handles client-side activities such as registration, profile management, requesting services, and tracking requests.
- **Provider Module** Manages provider functions including profile creation, browsing jobs, accepting requests, and completing tasks.
- **Admin Module** Allows the admin to monitor the system, manage users and providers, and oversee service requests.
- **Authentication Module** Responsible for user signup, login, and email verification to ensure secure access.
- **Service Request Module** Handles creation, submission, and management of service requests by clients.

- **Job/Proposal Management Module** Manages job assignments, proposals, acceptance, and completion of services.
- **Communication Module** Enables messaging between clients and providers for better coordination

8.2 Use case Diagram

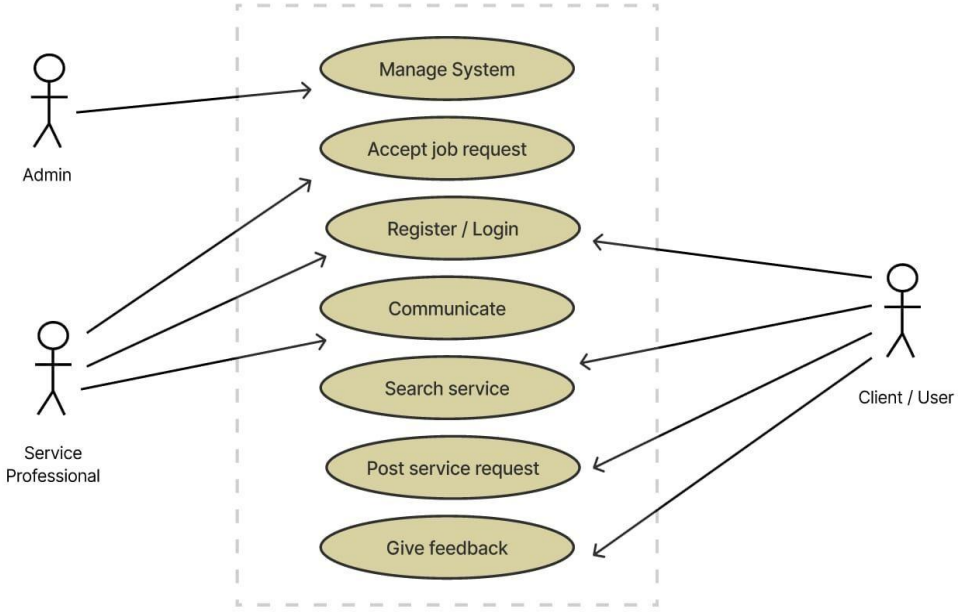


Figure 8.2

The use case diagram for the Koolie service platform shows the interaction between the Client/User, Service Professional, and Admin with the system. It highlights the main functionalities such as registration and login, searching for services, posting service requests, accepting job requests, communication between users and professionals, providing feedback, and system management by the admin. This diagram provides a clear overview of how different users interact with the system and how service requests are handled within the platform.

Based on their roles and access:

Actors and Use Cases

1. Admin

- Manage System

2. Service Professional

- Register / Login
- Accept Job Request
- Communicate

3. Client / User

- Register / Login
- Search Service
- Post Service Request
- Communicate
- Give Feedback

8.3 Data Flow Diagram

The data flow diagram (DFD) illustrates the flow of data within the system, showing how information moves between different components and processes. It consists of various symbols representing entities, processes, data stores, and data flows.

Level 0

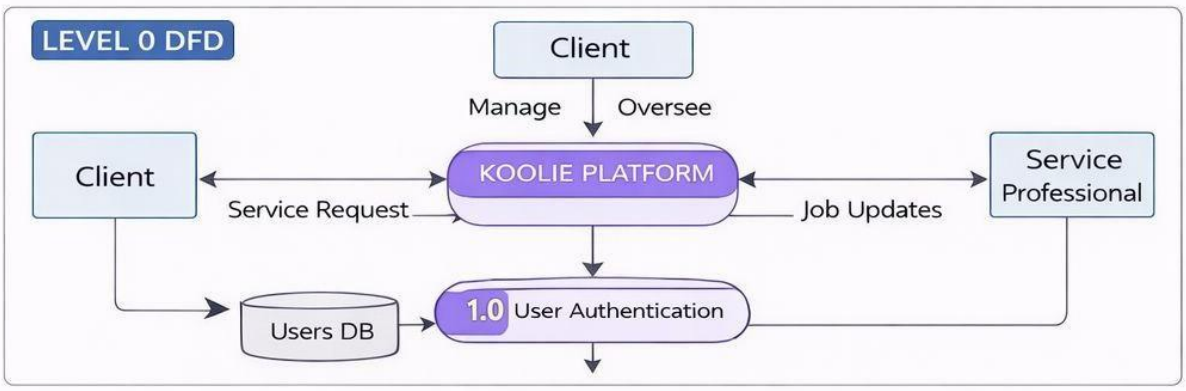


Figure 8.3.1

Level 1

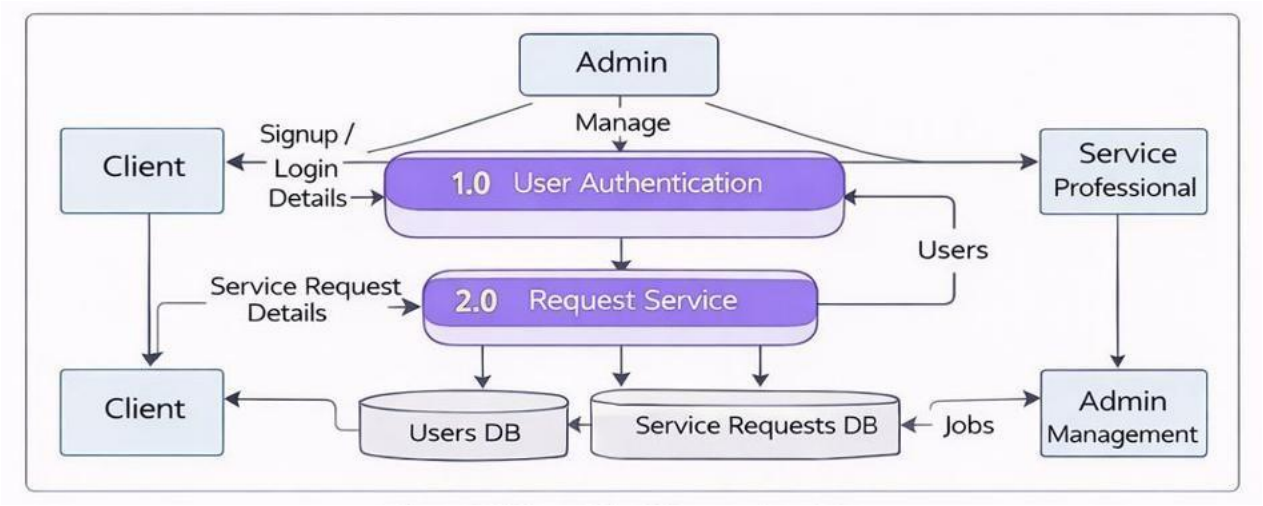


Figure 8.3.2

Level 2

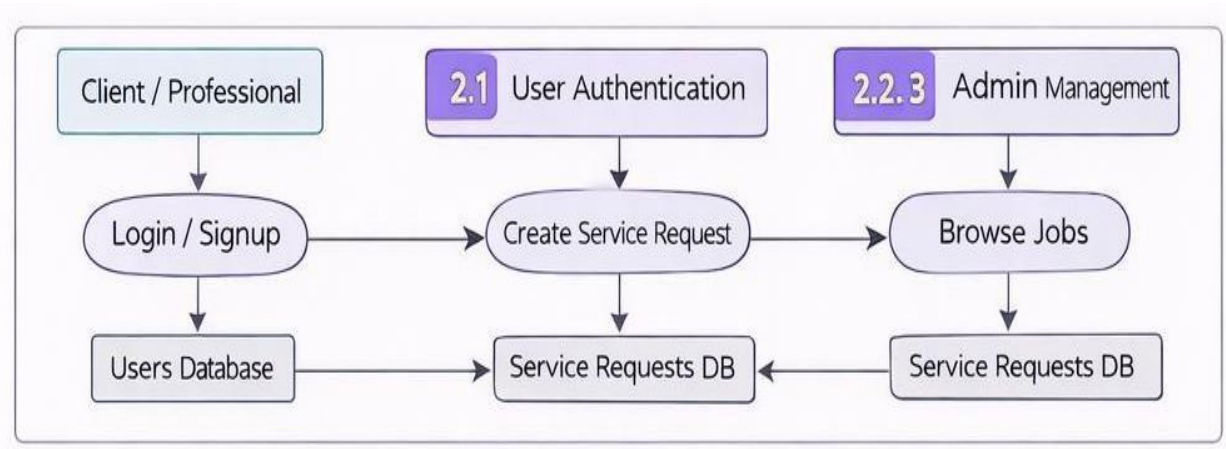


Figure 8.3.3

9. MODULAR DIVISIONS

9.1 Admin Module

The Admin Module acts as the control center of the platform and is responsible for managing and monitoring the overall system operations. Administrators have the authority to manage all user and service provider accounts to ensure proper platform usage. They can review, approve, or reject service listings submitted by providers to maintain service quality and reliability. The admin also manages financial operations, including handling payments, processing refunds, and managing payouts to service providers. In addition, the admin can view reports and platform-wide statistics to analyze system performance and user activity. This module also allows administrators to resolve disputes between users and providers and moderate content to maintain a safe and trustworthy platform environment.

9.2 User Module

The User Module is designed for customers who wish to browse and book services through the platform. Users can easily search for and explore different services based on their needs. The system allows them to book appointments with service providers and also provides options to reschedule or cancel bookings when necessary Users can also track the status of their requested service in real time. After receiving a service, users have the option to leave reviews and ratings, which helps maintain transparency and assists other users in making informed decisions.

9.3 Service Provider Module

The Service Provider Module is designed for professionals who offer services on the platform. Service providers can create their own profiles and list the services they offer, allowing users to discover and book them. Providers can manage their schedules and availability to ensure efficient time management. When users make booking requests, providers have the option to accept or reject them based on their availability. The module also enables providers to track their earnings and receive payouts for completed services. Additionally, service providers can respond to user reviews and ratings, which helps build credibility and improve service quality. They can also run promotions or special offers to attract more customers.

10. CODING PROCEDURE

10.1 Front End (React)

The front end of the **Koolie platform** is developed using React.js to provide a dynamic, responsive, and user-friendly interface. The application is structured using reusable components, enabling efficient development and maintainability. Key components include Login, Signup, Dashboard, Client Panel, Provider Panel, Service Request Form, Chat Interface, Profile Completion Page, and Admin Dashboard.

React hooks such as useState and useEffect are used to manage application state and handle dynamic data updates. Routing between different pages such as login, dashboard, and chat is implemented using React Router, ensuring smooth navigation without page reloads. The interface is designed to be intuitive, allowing users to easily create service requests, browse jobs, and communicate within the platform.

Structured forms are used to collect user inputs such as service details, provider skills, and location information including state and district. The frontend also integrates real-time features such as chat using Firestore listeners, enabling instant message updates between users. Overall, the React-based frontend ensures responsiveness, modularity, and an improved user experience.

10.2 Back End (Firebase)

The backend of the **Koolie system** is implemented using Firebase, which provides Backendas-a-Service (BaaS) capabilities. Firebase handles authentication, database operations, and real-time communication without requiring a separate server.

User authentication is managed using Firebase Authentication, supporting email/password login, email verification, and Google sign-in. The application uses Cloud Firestore as the primary database to store user profiles, service requests, chat messages, and role-based data.

Service requests are stored in structured documents containing fields such as service type, description, status, and location. Provider profiles include skill sets, experience, availability status, and location details. Real-time communication between clients and service providers is

achieved using Firestore's real-time listeners (onSnapshot), enabling instant updates in the chat module.

The backend also supports role-based access control, ensuring that clients, providers, and administrators can access only their respective functionalities. Firebase ensures secure data handling, scalability, and seamless integration between frontend and backend components.

10.3 Session Management

Session management in the **Koolie platform** is handled using Firebase Authentication. Once a user successfully logs in, Firebase maintains the authentication state across the application, allowing users to access protected routes such as dashboards and chat modules without repeated login.

The system uses authentication state listeners to track user sessions and automatically redirect users based on their login status and roles. For example, users are directed to client, provider, or admin dashboards depending on their assigned roles stored in Firestore.

When a user logs out, the session is securely terminated using Firebase's sign-out functionality, preventing unauthorized access. This approach ensures secure session handling and controlled access to system features.

10.4 Admin Module

An Admin module is implemented to manage and monitor system activities in the **Koolie platform**. Access to the Admin Dashboard is restricted to users with administrative privileges, verified through role-based authentication.

The admin can view and manage registered users, including service providers and clients. The module allows monitoring of all service requests, including their status (open, assigned, completed), and provides an overview of platform activity.

The admin panel helps in maintaining system transparency and enables centralized control over the platform. It ensures that the system operates efficiently by allowing monitoring of user activity, service requests, and overall platform performance.

11. TESTING AND IMPLEMENTATION

System testing is an essential phase in software development that ensures the developed system works correctly and meets all specified requirements. It focuses on identifying errors and verifying that the system performs as expected under different conditions. Testing helps improve the reliability, accuracy, and overall quality of the application.

In the Koolie web application, testing was conducted to verify the proper functioning of all core modules — including user authentication, session management, real-time data operations, role-based access control, and the overall user interface. The system was tested using multiple user accounts and data scenarios to ensure that authentication flows are secure, Firestore data operations are accurate, and the admin access controls function correctly under all conditions. Both functional and structural testing approaches were used to validate system performance.

TEST CHARACTERISTICS

1. Effective Error Detection
2. Non-Redundancy
3. It should meet system objectives.
4. It should be simple and effective

BLACK BOX TESTING

Black box testing checks system functionality based on input
and output.

It was used to test:

- User Authentication Flows
- Route Access Behaviour
- Data Input and Form Submission
- Real-Time UI Updates

WHITE BOX TESTING

White box testing focuses on internal logic.

It ensured:

- Firebase Initialisation Logic
- onAuthStateChanged Observer
- ProtectedRoute Logic
- Firestore CRUD Functions
- useUserRole Hook

UNIT TESTING

Each module was tested separately.

- Authentication Module
- AuthContext Provider
- Firestore Service Functions
- useUserRole Hook
- ProtectedRoute Component
- UI Components

INTEGRATION TESTING

Ensured all modules work together:

- Authentication → Session → Route Access
- Login → Role Detection → UI Rendering
- Data Input → Firestore Write → Real-Time UI Update
- Update Operation → Firestore → Live Refresh
- Logout → Session Clearance → Route Protection
- Frontend → Firestore Security Rules → Backend Enforcement

VALIDATION TESTING

Validation testing confirms that the fully integrated system meets the original requirements and performs correctly in realistic usage conditions.

11.1Home Page

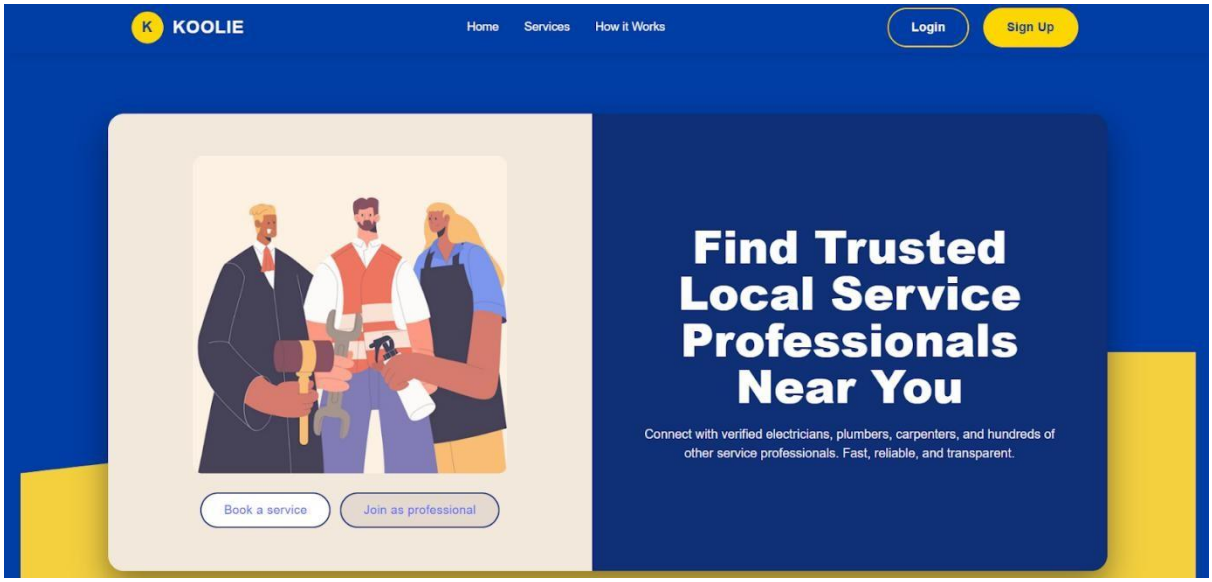


Figure 11.1

11.2 Signup Page

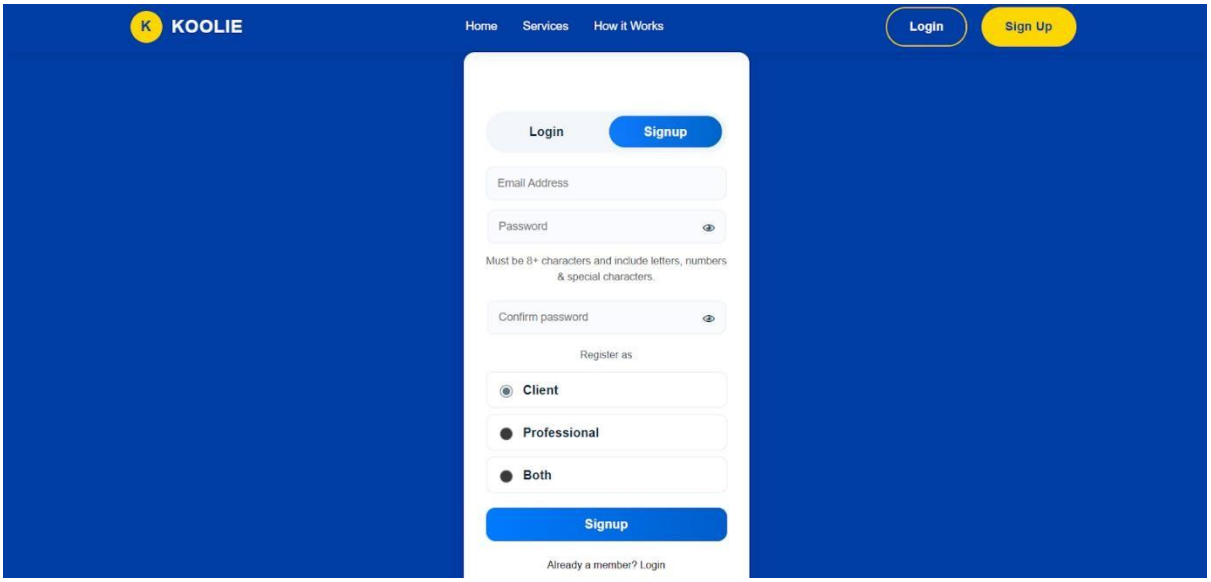


Figure 11.2

11.3 Client Dashboard

11.3.1 Request a Service

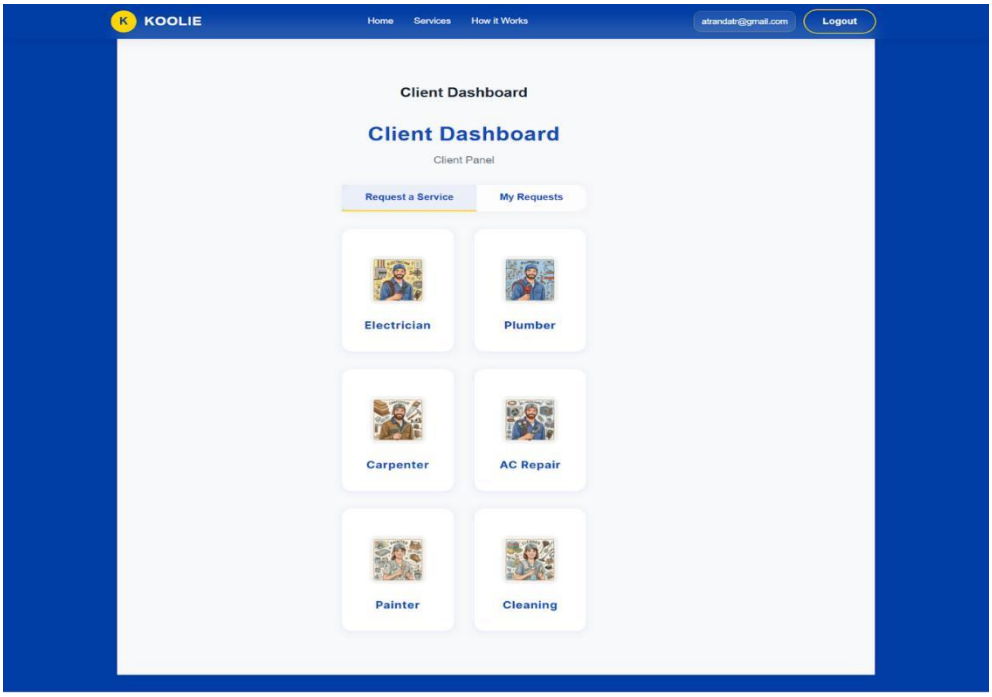


Figure 11.3.1

11.3.2 My Request

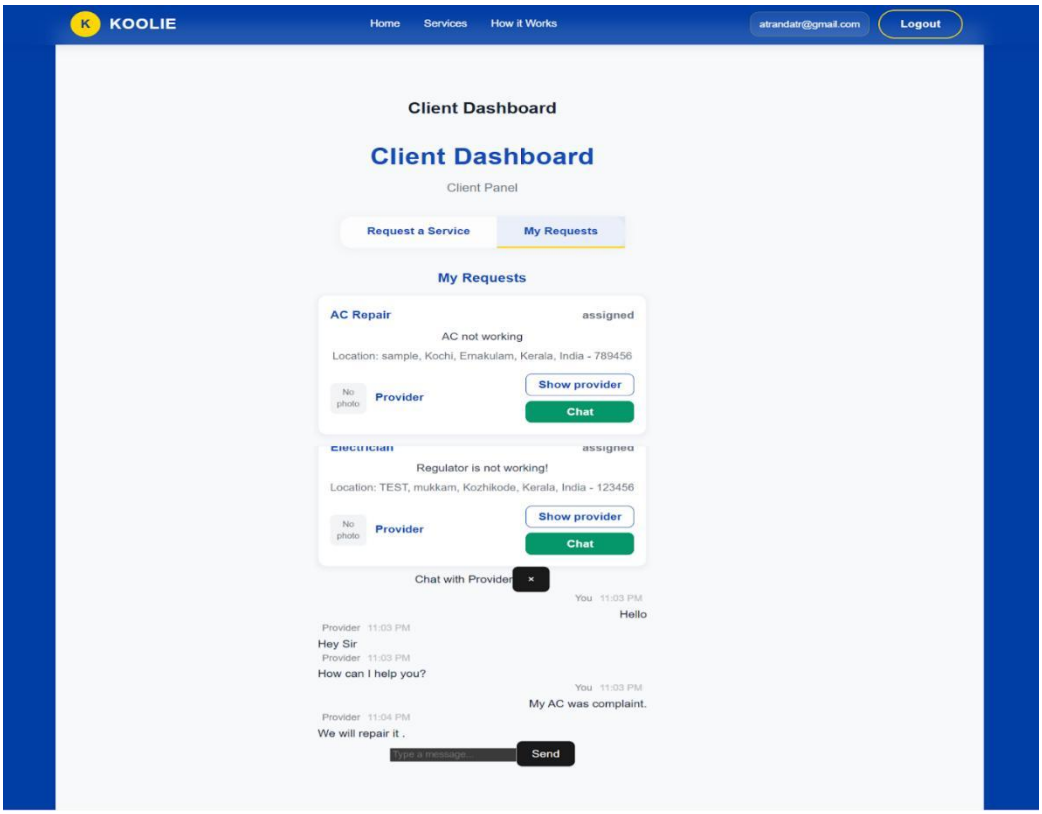


Figure 11.3.2

11.4 Service Provider Dashboard

11.4.1 Browse Jobs

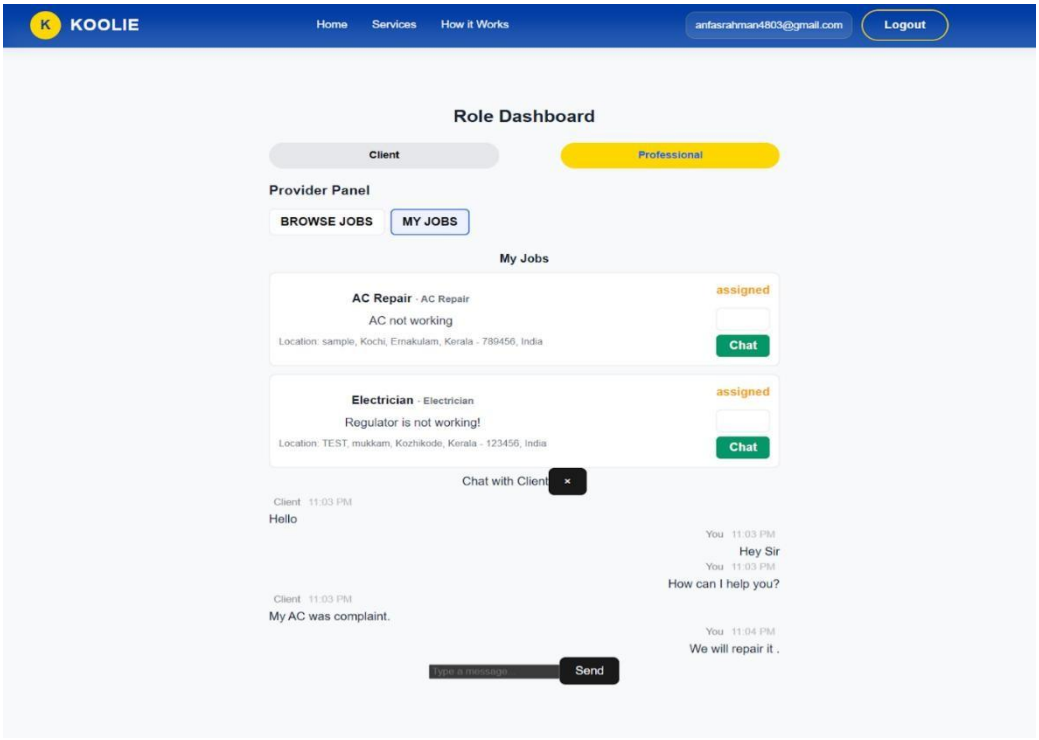


Figure 11.4.1

11.4.2 My jobs

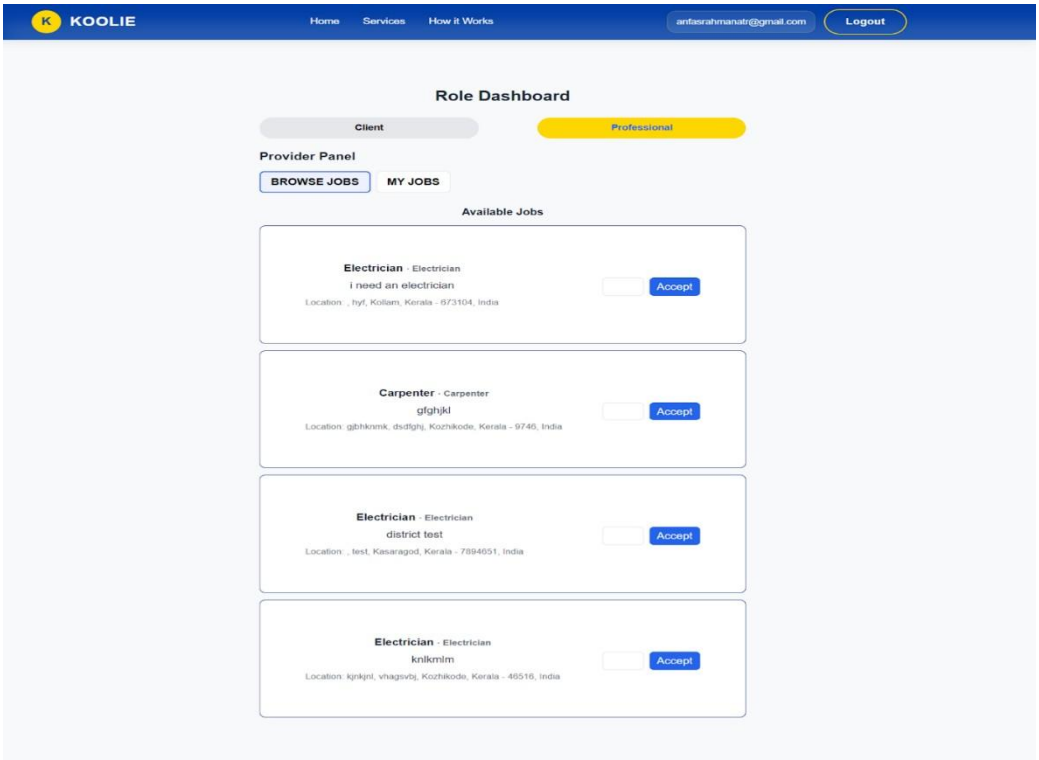


Figure 11.4.2

Koolie0 > client > src > pages > Login.jsx > ...

```jsx
function Login() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const [showPassword, setShowPassword] = useState(false);
  const [error, setError] = useState(null);
  const navigate = useNavigate();

  // reuse same strength rule as signup
  const isStrongPassword = (p) => {
    return /^(?=.*[A-Za-z])(?=.*\d)(?=.*[^A-Za-z\d]).{8,}$/.test(p);
  };

  const handleSubmit = async (e) => {
    e.preventDefault();
    setError(null);
    if (!isStrongPassword(password)) {
      setError(
        "Password must be at least 8 characters and include letters, numbers, and special characters."
      );
      return;
    }
    try {
      const cred = await signInWithEmailAndPassword(auth, email, password);
      const user = cred.user;
      if (!user.emailVerified) {
        setError("Please verify your email before logging in. Check your inbox for the verification email.");
        await signOut(auth);
        return;
      }
      const uid = user.uid;

      const userRef = doc(db, "users", uid);
      const snap = await getDoc(userRef);
      if (!snap.exists()) {
        // create a minimal user doc for first-time sign-ins
        try {
          await setDoc(userRef, {
            email: user.email || email,
            roles: { client: true, provider: false },
            profileCompleted: false,
            clientProfile: {},
            providerProfile: {},
            skills: [],
            createdAt: serverTimestamp(),
          });
```

Figure 11.5 1

Koolie0 > client > src > App.jsx > ...

```jsx

function HomePage() {
  return (
    <>
      {/* keep spacing so content isn't hidden by fixed navbar */}
      <div style={{ paddingTop: "92px" }}>
        {/* Home component with hero and other sections */}
        <Home />
      </div>
    </>
  );
}

function App() {
  return (
    <AuthProvider>
      <Router>
        <Navbar />
        <Routes>
          <Route path="/" element={<HomePage />} />
          <Route path="/login" element={<Login />} />
          <Route path="/signup" element={<Signup />} />
          <Route path="/complete-profile" element={<CompleteProfile />} />
          <Route path="/role-dashboard" element={<RoleDashboard />} />
          <Route path="/client-dashboard" element={<ClientDashboard />} />
          <Route path="/request" element={<RequestForm />} />
          <Route path="/provider-dashboard" element={<ProviderDashboard />} />
          <Route path="/admin-login" element={<AdminLogin />} />
          <Route path="/admin" element={<AdminDashboard />} />
        </Routes>
      </Router>
    </AuthProvider>
  );
}

export default App;
```

Figure 11.5 2

Koolie0 > client > src > contexts > AuthContext.jsx > ...

```jsx
import React, { createContext, useContext, useEffect, useState } from 'react';
import { onAuthStateChanged } from 'firebase/auth';
import { auth } from '../services/firebase';

const AuthContext = createContext();

export const useAuth = () => {
  const context = useContext(AuthContext);
  if (!context) {
    throw new Error('useAuth must be used within an AuthProvider');
  }
  return context;
};

export const AuthProvider = ({ children }) => {
  const [user, setUser] = useState(null);
  const [loading, setLoading] = useState(true);

  useEffect(() => {
    const unsubscribe = onAuthStateChanged(auth, (user) => {
      setUser(user);
      setLoading(false);
    });

    return unsubscribe;
  }, []);

  const logout = async () => {
    await auth.signOut();
  };

  const value = {
    user,
    logout,
    loading,
  };

  return (
    <AuthContext.Provider value={value}>
      {children}
    </AuthContext.Provider>
  );
};
```

Figure 11.5 3

12.RESULTS AND DISCUSSION

The proposed Koolie successfully implements a structured system for connecting clients with local service professionals such as electricians, plumbers, and carpenters through a webbased application. The system was developed using React for the frontend and Firebase for backend services including authentication, database management, and real-time communication.The system was tested with multiple user roles including clients, service providers, and administrators. Users were able to register, log in using email/password or Google authentication, and complete their profiles with relevant details such as skills, experience, and location. The authentication process, including email verification, ensured secure access to the platform.

The service request module allows clients to create requests by specifying service type, description, and structured location details such as state and district. The system effectively stores and manages service requests in the Firestore database. Service providers can browse available jobs filtered based on their registered location, particularly district-based matching, which improves the relevance of job listings.The job assignment mechanism allows service providers to accept requests, updating the request status from open to assigned. Once a job is assigned, both client and provider gain access to a real-time chat feature, enabling direct communication within the platform. The chat system, implemented using Firestore real-time listeners, provides instant message updates between users.

The platform also includes an admin module that allows monitoring of registered users and service requests, providing administrative control over the system. The user interface was designed to be simple and intuitive, allowing users to navigate dashboards, create requests, browse jobs, and communicate effectively.Overall, the developed system demonstrates reliability, scalability, and usability. It provides a practical solution for connecting service seekers with providers while supporting real-time interaction and location-based matching, making it an efficient platform for everyday service management.

13. CONCLUSION AND FUTURE SCOPE

The proposed Skill-Based Project Grouping and Monitoring System effectively addresses the challenges associated with traditional manual project group formation and supervision in academic institutions. By introducing an automated, skill-based approach, the system ensures balanced team formation based on students' skills, interests, and experience levels, thereby improving collaboration and overall project quality.

The system provides structured project monitoring through real-time tracking of milestones, submissions, and progress. With secure user authentication, role-based access control, centralized data management, and continuous feedback mechanisms, the platform ensures transparency, accountability, and efficient supervision. The reduction in faculty workload and improvement in coordination among students and guides demonstrate the reliability and usability of the system. Overall, the project delivers a scalable, efficient, and user-friendly solution that enhances the effectiveness of projectbased learning.

Future Scope

The system can be further enhanced by incorporating advanced analytics and intelligent recommendation techniques to improve group formation accuracy. Future improvements may include the integration of AI-based skill analysis, performance analytics dashboards, and automated evaluation support for faculty members.

Additional enhancements could involve institution-wide deployment, integration with existing academic management systems, and support for industry-sponsored projects. Expanding the system with mobile notifications, detailed progress analytics, and longterm performance tracking can further improve usability and impact. These advancements will make the platform a comprehensive and intelligent solution for academic project management and monitoring.

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